

CSA1717-ARTIFICIAL INTELLIGENCE

CO2-AT3-DRY RUN TEST

PROBLEM STATEMENT

Question 1:

Problem Statement

Consider the following weighted graph in which each edge represents the cost of moving from one node to another. The heuristic values $h(n)$ indicate the estimated cost from each node to the goal node **G**.

(A)

/ \

2 / \ 4

/ \

(B)---3---(C)

| \ |

7 | \ 2 | 3

| \ |

(D)---2---(E)

\

\ 2

(G)

Heuristic Values

Node $h(n)$

A 7

B 6

C 4

D 3

Node $h(n)$

E 2

G 0

Task

Perform a dry run of the **A* Search Algorithm** to find the shortest path from the **Start Node (A)** to the **Goal Node (G)**.

For each iteration, display:

- Current Node
- Open List
- Closed List
- Actual Path Cost **$g(n)$**
- Heuristic Value **$h(n)$**
- Evaluation Function **$f(n) = g(n) + h(n)$**

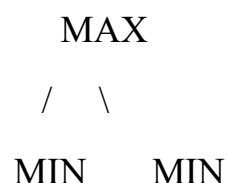
Finally,

- Determine the optimal path from **A** to **G**.
- Calculate the total path cost.
- Explain how the heuristic values influence the search process and help in selecting the most promising node at each step.

Question 2:

Problem Statement

Consider the following two-player game tree. The root node represents the **MAX** player, while the second level represents the **MIN** player. The leaf nodes contain the utility values that determine the outcome of the game.



```

      / | \   / | \
     3  5  6 9  1  2

```

Task

Perform a dry run of the **Minimax Algorithm with Alpha-Beta Pruning** by evaluating the game tree from **left to right**.

For each node, display:

- Alpha (α) value
- Beta (β) value
- Selected utility value
- Pruned node(s), if any

Finally,

- Determine the best move for the **MAX** player.
- Find the final minimax value.
- Identify the node(s) that are pruned using Alpha-Beta Pruning.
- Explain how Alpha-Beta Pruning reduces the number of node evaluations without affecting the optimal decision.